TankGuard

Technical Specification & Invention Disclosure Summary

Document Ref: TG-SPEC-2026-001 · October 2026 · IoT Water Tank Monitoring & Safe Pump-Control System

1. Title of Invention

A Self-Powered IoT System for In-Tank Water Quality Monitoring and Safe Pump Control for Domestic and Institutional Water Storage Tanks.

Full title: *"A self-powered IoT-based in-tank water quality monitoring system with edge-computed composite scoring, trend prediction and a local-network dashboard, paired with a mains-isolated home pump controller for domestic and institutional water storage tanks."*

2. Abstract

Water storage tanks in homes, schools, hospitals and apartment buildings degrade silently: disinfectant residual dissipates within 24–48 hours, suspended particles settle into a sludge layer at the tank floor, biofilm forms on internal surfaces, and algae proliferate in warm conditions. By the time degradation is detected by smell, taste or sight, the water is already unsafe. Existing solutions either monitor without acting, or clean by draining and scrubbing the entire tank — wasting water and labour.

TankGuard is a single, self-powered IoT unit that is mounted on the tank lid and continuously measures five water-quality parameters — **turbidity, TDS, pH, temperature and water level** — through submerged probes. An on-device ESP32 microcontroller performs all computation at the network edge: it fuses the readings into a weighted composite **TankScore**, analyses 14-day trends, and classifies the contamination event (gradual biofilm growth, source-water contamination, or sludge resuspension). When the score indicates that the water needs attention, the system raises a local alert and can switch the pump through a separate home unit using an opto-isolated relay and a motor-rated contactor — the ESP8266 never touches mains. A crash or lost link forces the pump OFF. All logic runs locally with no cloud, subscription or internet requirement; a Wi-Fi access point and embedded web server present an interactive dashboard to any phone or browser on the local network. The tank unit is powered by a small solar panel and battery, with duty-cycled sensing for multi-week autonomy.

3. Components Used

TankGuard is built entirely from commercially available off-the-shelf (COTS) components. The table below lists the components of a working embodiment, with a representative part and its function.

Component	Representative Part	Interface	Function
Microcontroller (edge processor)	ESP32-WROOM-32 DevKit	Wi-Fi, GPIO, I²C, 1-Wire, ADC	Runs all firmware — sampling, TankScore computation, trend regression, event classification, embedded HTTP server, mDNS, flash logging and deep-sleep power management
ADC	ADS1115, 16-bit, 4-channel	I²C @ 0x48	High-resolution digitisation of the pH electrode and analog sensors; ESP32's internal ADC is too coarse for millivolt-level pH
Turbidity sensor	DFRobot SEN0189 (or generic clone)	Analog	90° infrared nephelometry — primary indicator of suspended solids, biofilm fragments and sludge resuspension (40% of TankScore)
TDS sensor	Gravity SEN0244 / XKC-Y25-NPN	Analog + NTC	Two-electrode conductivity with temperature compensation — dissolved-solids leaching from settled debris (25% of TankScore)
pH sensor	Gravity SEN0161-V2 / glass electrode + front end	Analog via ADS1115	Potentiometric pH — indicates algal activity, anaerobic decay and corrosion (20% of TankScore)
Temperature probe	DS18B20 waterproof (stainless)	1-Wire	Temperature compensation plus independent biological-activity indicator (5% of TankScore)
Water-level sensor	JSN-SR04T waterproof ultrasonic	Trigger / Echo GPIO	Water volume and refill detection; enables discrimination of source contamination from sludge resuspension (10% of TankScore)
Home pump-control unit	ESP8266 + 1.8″ ST7735 TFT + opto-relay + AC-3 contactor	Wi-Fi / nRF24L01+; relay contact → contactor coil	Separate indoor unit that displays tank status and switches the pump safely; the ESP8266 never touches mains
Power system	10 W solar panel, 4×18650 Li-ion, CN3791 MPPT, LM2596 buck + AMS1117	—	Self-sustaining off-grid power with 2–3 weeks of zero-sun autonomy; regulated 12 V / 5 V / 3.3 V rails
Enclosure	IP66 ABS/polycarbonate box + IP68 cable glands + heat shield	—	Weatherproof, heat-protected housing mounted on the tank lid
Calibration set	pH 4.0 / 7.0 buffer sachets + distilled water + KCl storage solution	—	Two-point pH calibration and probe conditioning
Firmware	ESP-IDF / Arduino framework + LittleFS	Wi-Fi AP/STA	All edge logic and the local dashboard (software-implemented portion)

4. Features

- **Multi-parameter sensing** — turbidity, TDS, pH, temperature and water level measured continuously in-tank
- **TankScore composite index** — one 0–100 number answering "is it time to clean?"
- **Trend-predictive analysis** — 14-day linear regression detects degradation before visible symptoms
- **Event classification** — distinguishes biofilm growth, source-water contamination and sludge resuspension
- **Mains-safe pump control** — opto-relay → contactor → pump, with fail-safe OFF
- **Manual override** — one physical button plus dashboard "Tank Cleaned" confirmation
- **Solar-powered autonomy** — runs off-grid for weeks without sun
- **Cloud-free local dashboard** — any phone/browser, no internet, no account
- **Edge data logging** — full history stored on-device; optional OTA updates and alert exports

5. What Makes the System Unique

The invention is distinguished from prior art (sensor-only monitors, full-tank drain cleaners, industrial camera-based sludge systems, in-pipe point-of-use devices) by the following non-obvious combinations:

Unique Aspect	Distinguishing Feature
1. Trend-predictive TankScore	Triggering uses a weighted composite with 14-day slope and event classification, not a single-sensor threshold — enabling action *before* visible degradation.
2. Sludge-resuspension detection	Correlating turbidity spikes against level changes distinguishes resuspension from source contamination — a specific diagnosis absent from prior art.
3. Edge-only IoT architecture	All sensing, scoring, decision and alerting execute on-device; the dashboard is self-hosted over the local network. No cloud, subscription or account — a deliberate architectural contrast to all surveyed IoT monitors.
4. Duty-cycled solar autonomy	Deep-sleep MCU with switched sensor power rails achieves multi-week off-grid operation while maintaining continuous logical monitoring.
5. Single integrated sensing unit	Sensing, edge computing, power and user interface combined in one tank-lid-mounted device; no tank penetration or plumbing connection is required.

6. Expected Cost

Two build tiers are available. All figures are parts-only and exclude GST, freight and installation.

Tier	Description	Parts Cost	Installed (approx.)
Full / proposal build	Grade-A Gravity sensors, IP66 enclosure, home pump-control unit, full integration	**₹16,580**	~₹17,500
Minimum prototype	Generic/clone sensors, salvaged 18650 cells, basic enclosure	**~₹6,500**	~₹7,200

6.1 Full Build Breakdown

Module M1 — Core Monitor		**₹15,500**
ESP32, ADS1115, sensors, solar/battery/power, enclosure, calibration, engineering	Five-parameter monitor with solar power, edge dashboard and TankScore logic	—
Module M2 — Home Pump Control		**₹1,080**
ESP8266, 1.8″ TFT, opto-relay, AC-3 contactor, enclosure, MCB	Indoor pump controller with local display, override button and fail-safe OFF	—
Total Parts Cost		**₹16,580**

Notes: The pump module is deliberately isolated: opto-relay → motor-rated contactor → pump, with the ESP8266 kept away from mains and the pump forced OFF on crash or link loss. Mains wiring must be done by a licensed electrician. The minimum tier (~₹6,500) trades sensor grade and enclosure quality for cost; readings are adequate for trend detection (±5–10%) but not for compliance-grade measurement.

7. Technical Specifications

7.1 Hardware Specifications

Parameter	Specification
Microcontroller	ESP32-WROOM-32 — dual-core Xtensa LX6 @ 240 MHz, 520 KB SRAM, 4 MB flash; Wi-Fi 802.11 b/g/n, Bluetooth LE
ADC	ADS1115 — 16-bit, 4-channel, I²C; programmable gain; internal reference
Turbidity	0–3000 NTU range (practical 0–100 NTU), IR 90° nephelometry, IP68 probe
TDS	0–1000 ppm (extendable), 3–5.5 V excitation, ±10% FS, temperature-compensated
pH	0–14 pH, glass electrode, two-point calibrated (4.0 / 7.0), ±0.1 pH after calibration
Temperature	−10 to +85 °C, ±0.5 °C (DS18B20), waterproof stainless probe
Water level	25–450 cm (JSN-SR04T), ±1 cm, IP67 ultrasonic
Pump interface	Opto-isolated 5 V relay (10 A / 250 V) → motor-rated AC-3 contactor; one local override button
Power — solar	10 W, 12 V polycrystalline panel (5 W 6 V in minimum tier)
Power — battery	4×18650 Li-ion (~10 Ah @ 3.7 V equivalent); 2–3 weeks zero-sun autonomy
Charging	CN3791 MPPT charge controller with over-charge / over-discharge protection
Regulated rails	12 V battery bus · 5 V (sensor excitation) · 3.3 V (MCU/logic) via LM2596 + AMS1117
Sleep current	< 1 mA in deep sleep between measurement cycles
Enclosure	IP66 polycarbonate/ABS, heat shield, IP68 cable glands; lid-mounted
Operating temperature	0–55 °C ambient (electronics); probes −10 to +85 °C

7.2 Software Specifications

Parameter	Specification
Firmware framework	ESP-IDF (or Arduino-ESP32); modular C/C++
Measurement cycle	Wake every 3 hours (configurable); median-filtered multi-sample acquisition
TankScore	0–100 weighted composite: turbidity 40%, TDS 25%, pH 20%, level 10%, temperature 5%
Trend analysis	Linear regression over 14-day window (112 samples); slope/acceleration penalties
Event classification	3 classes — gradual degradation, source-water contamination, sludge resuspension
State machine	GOOD (80–100) → WATCH (50–79) → ACTION ADVISED (30–49) → CLEAN REQUIRED (0–29)
Pump control	Auto ON below LOW level, OFF above HIGH level; manual override auto-releases after 10 min; pump OFF on link loss
Verification metrics	Four separate quantities, never interchangeable: turbidity reduction `R_T = (T_pre − T_post)/T_pre × 100` (%); estimated sediment reduction `R_S = (S_pre − S_post)/S_pre × 100` (%, from the NTU→mg/L calibration); TankScore change `ΔQ = Q_post − Q_pre` (points, positive = better); estimated sediment mass removed `m = (S_pre − S_post) × V` (g). The ratio form is valid only for reduction metrics — TankScore is higher-is-better and is reported as a point change, never as a percentage of its own baseline.
Local server	Embedded HTTP server + mDNS (`tankguard.local`), AP or STA mode
Dashboard	HTML5/JS single-page — gauges, trend charts (24 h / 7 d / 30 d / 90 d), TankScore banner, pump status, history
Storage	LittleFS on flash — ring buffer of samples + event log (pre/post turbidity, pre/post TankScore, and the derived R_T, R_S, ΔQ and m)
Connectivity	Wi-Fi only by default (no cloud); optional MQTT/e-mail/SMS alerts are notification-only, never control
OTA	Optional over-the-air firmware update while installed
Power management	Deep-sleep scheduler with switched sensor power rail; <1 mA idle

8. Drawings

The following figures are schematic and not to scale. They illustrate the architecture, operating cycle, control states and physical mounting of the invention.

Fig. 1 — System Architecture (five-layer IoT stack)

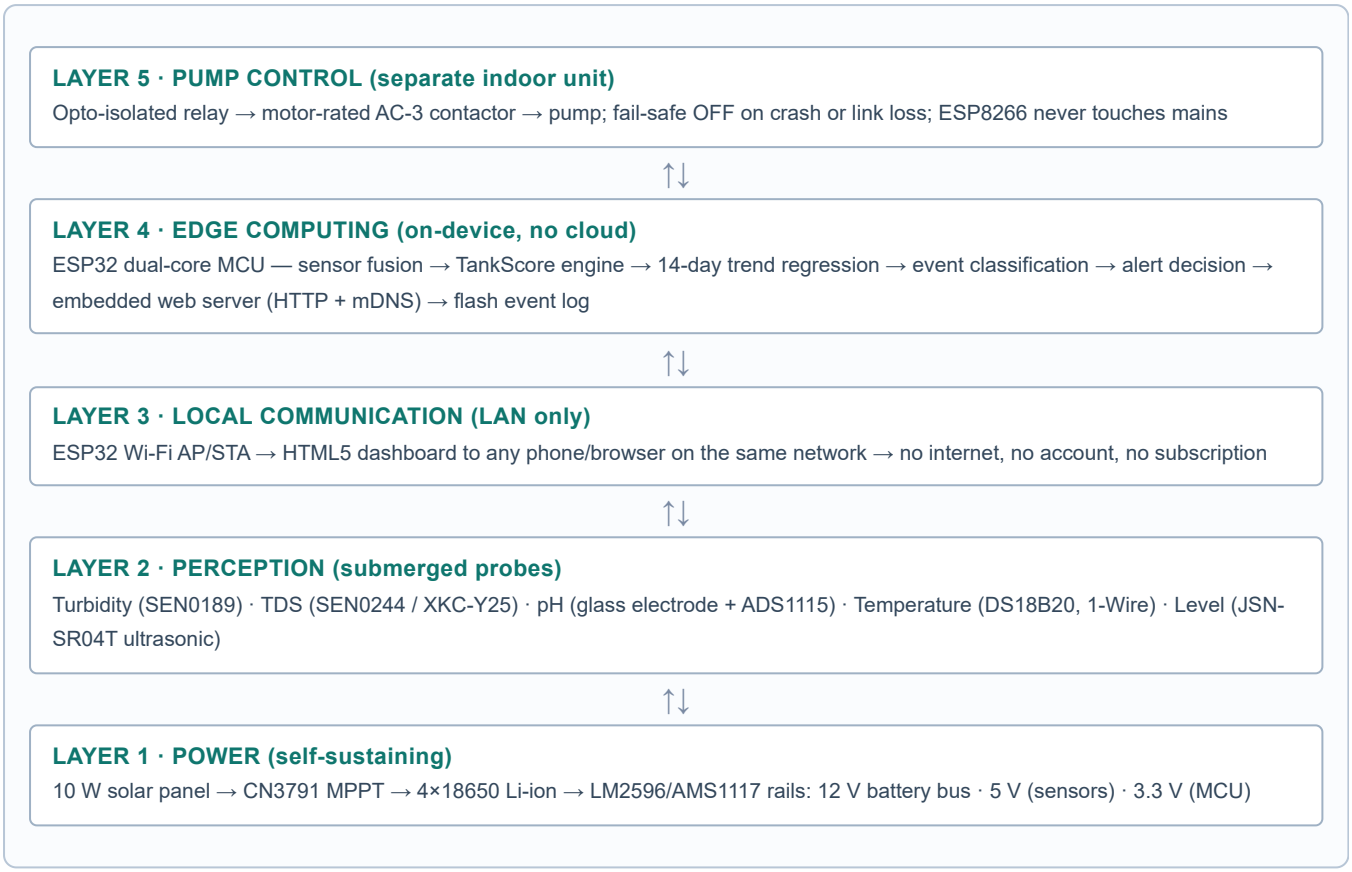


Fig. 1 — Four functional layers plus power; all computation stays at Layer 4 (edge).

Fig. 2 — Operating Cycle (sense → compute → actuate → verify)

Step 1 — Wake: RTC timer wakes ESP32 (every 3 h); sensor power rail energised.

Step 2 — Acquire: Read turbidity, TDS, pH, temperature, level; median-filter each.

Step 3 — Score: Normalise + weight parameters into TankScore; append sample to 14-day buffer.

Step 4 — Analyse: 14-day regression + classify event (degradation / source contamination / sludge resuspension).

Step 5 — Decide: Map score to GOOD / WATCH / ACTION ADVISED / CLEAN REQUIRED; raise a local alert.

Step 6 — Control: Home unit runs the pump via opto-relay → contactor, or leaves it OFF; fail-safe OFF on crash or link loss.

Step 7 — Report: Log event; update local dashboard (readings, TankScore, trends, pump status, battery).

Step 8 — Sleep: De-energise sensors; deep sleep <1 mA until next cycle.

Fig. 2 — Edge scoring and alerting; pump control is isolated in a separate home unit.

Fig. 3 — TankScore State Diagram



GOOD → WATCH → ACTION ADVISED → CLEAN REQUIRED | after cleaning and re-measurement, the score recovers toward GOOD; otherwise the alert persists.

Fig. 3 — TankScore state transitions and corresponding system actions.

Fig. 4 — Mounting Schematic (elevation, not to scale)

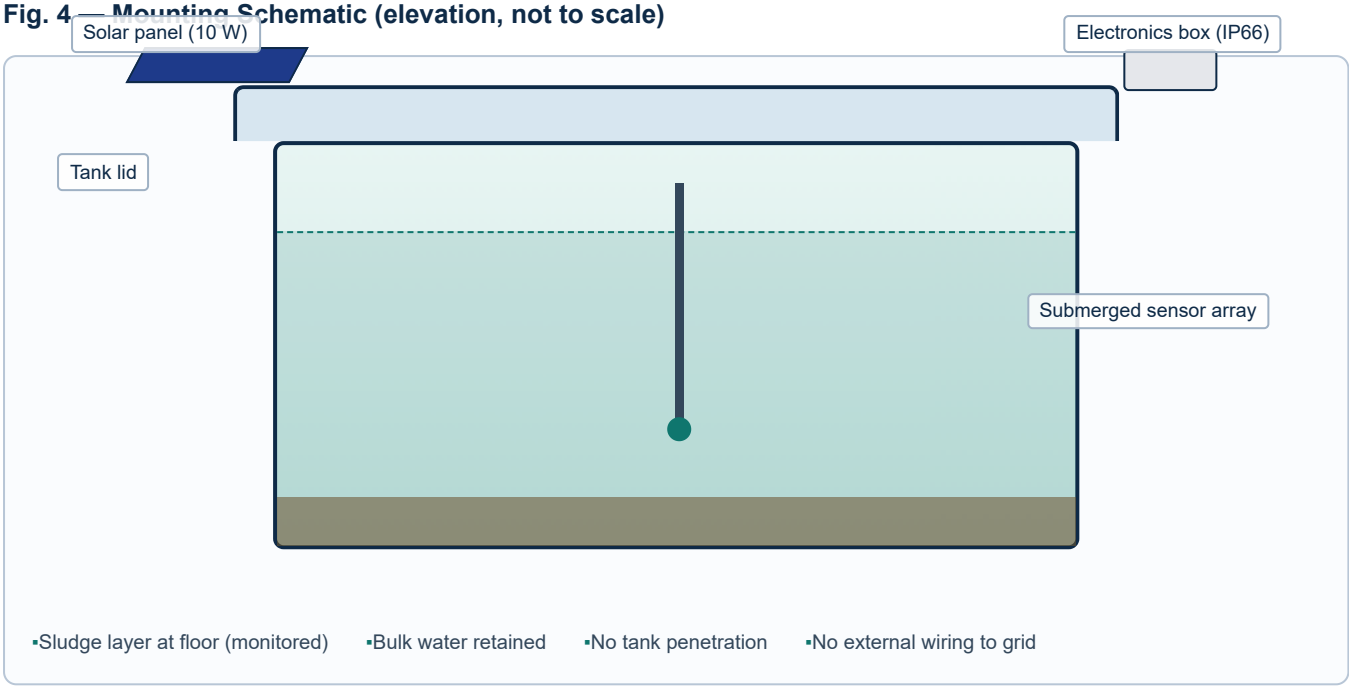


Fig. 4 — Lid-mounted unit with suspended probes; no plumbing connection is required.

Summary: TankGuard converts water-tank maintenance from calendar-based guesswork into evidence-based care — sensing at the edge, scoring locally, alerting early, and controlling the pump safely, with no cloud dependency.